NLP Project 1: Preliminary Contribution Statement

COMS4054A / COMS7062A / COMS7066A • 14 October 2026

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Project: *Interpretability of Model Collapse* (working title: *Collapse around the circuit: induction heads under recursive training*)

Research question. A small transformer learns the induction-head symbol task of Singh et al. (2024) and is then trained recursively on data generated by its previous generation (Shumailov et al., 2024). Does collapse degrade the induction circuit itself, or only what the circuit is asked to produce? How does the fraction of real data mixed into each generation change this, in both the base and the extended version of the task?

Division of work

Each member contributes one third (33.3%) of the project.

Member	Share	Experiments	Writing and admin
Sunay	33.3%	Core implementation and collapse pipeline; A+C sweeps and hyper-parameter selection; temperature test and poison-pulse runs	Method section; project outline and change plan
Matthew	33.3%	Single-generation analysis: training curves, attention maps, embeddings, convergence and loss floor; extra logging; headline figure	Introduction, Background and Results sections; extended abstract LaTeX; Faculty of Science AI ethics statement
Harshil	33.3%	Inheritance-law fit; gate audit and statistics; finite-data control run; mechanism analysis of the hallucination	Discussion and limitations; NeurIPS 2024 checklist; README.txt and requirements.txt

Sunay Master

- Build the study→query task generator (base and extended), the 2-layer transformer, the training loop, and the collapse pipeline (real/synthetic mixing, convergence gate with retries, collision-free seeding).
- Build the circuit probes (previous-token and induction scores, causal head ablation), conditional diversity and off-context mass; run the full sweep over real-data fractions, seeds and both modes; select hyper-parameters.
- Run the pre-registered temperature test of the critical real-data fraction and the poison-pulse experiment, and write the Method section.

Matthew Purkiss

- Produce the single-generation training curves, attention-map and embedding figures, and check that extended mode converges and reaches its predicted loss floor.
- Add validation loss and all-query perplexity to the logging, add mean-ablation to the probes, and build the composite headline figure.

- Write the extended abstract (Moodle template) including Introduction, Background and Results, and the Faculty of Science AI ethics statement.

Harshil Vallabh

- Fit the recursion model to the saved results, with held-out fractions as a test, and audit which generations fail a stricter convergence gate.
- Run the control that gives generation 0 the same data budget as later generations, and analyse which heads produce the off-context (hallucinated) probability.
- Write the Discussion and limitations, the NeurIPS 2024 checklist, README.txt and requirements.txt, and check that every number in the abstract traces to a file.

Shared by all members

- The two-page extended abstract (PDF), the full code, and checking every reference.
- The final contribution statement, which will report any changes to this plan and why.

Agreement

All group members have read and agree to this division of work.

Sunay Master	Matthew Purkiss	Harshil Vallabh
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____